Lemmas not found in EGMO

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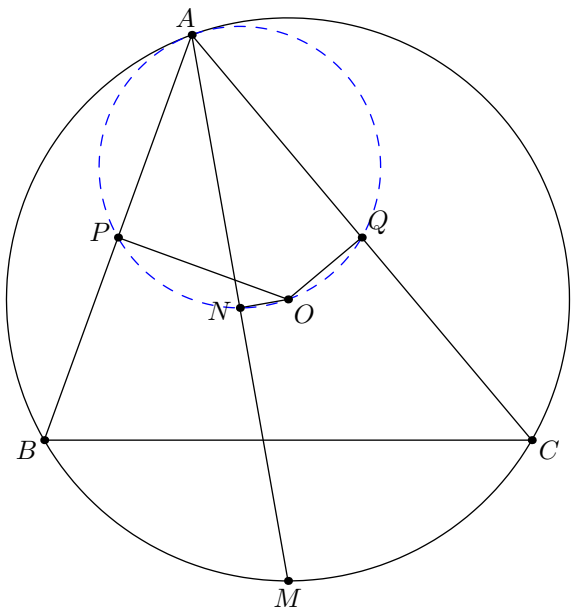
This is a collection of lemmas, or nice problems, that are recurring enough to warrant a handout. Note that I am including lemmas that I haven't seen in EGMO or any of Evan's Handouts, which can be found here: https://web.evanchen.cc/olympiad.html.

§1 Symmetry of angle bisector

The main claim here:



Lemma 1.1 (Angle bisector reflections)

In a triangle $\triangle ABC$, let O be the circumcenter. Also, define M as the midpoint of arc BC not containing A. Let the midpoint of line AM be N. Then the perpendicular bisectors of AB and AC are reflections over line ON.

Proof. Try to do it yourself! It should be intuitive.

Notice that $APNOQ$ is cyclic since it is just (ABC) under a homothety of factor $1/2$ centered at A. But then it follows that

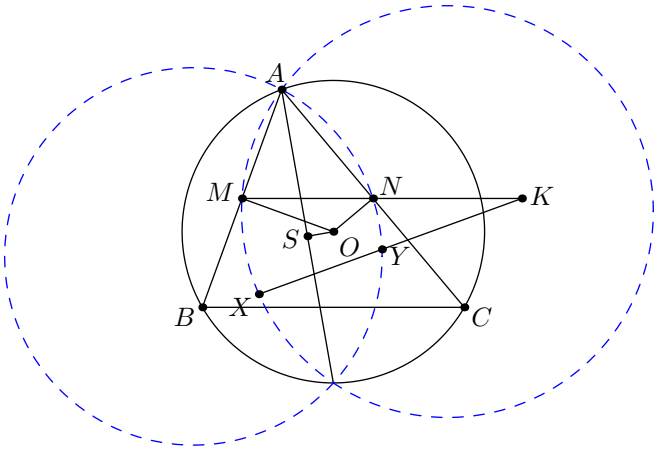
$$\measuredangle PON = \measuredangle PAN = \measuredangle NAQ = \measuredangle NOQ$$

which implies the conclusion. □

The most prominent example I can think of is Shortlist 2013 G2:

Example 1.2 (Shortlist 2013 G2)

Let ω be the circumcircle of a triangle ABC. Denote by M and N the midpoints of the sides AB and AC, respectively, and denote by T the midpoint of the arc BC of ω not containing A. The circumcircles of the triangles AMT and ANT intersect the perpendicular bisectors of AC and AB at points X and Y, respectively; assume that X and Y lie inside the triangle ABC. The lines MN and XY intersect at K. Prove that $KA = KT$.



Proof. I actually claim that X and Y are the reflections of M and N across OS, where S is the midpoint of AM (hint: use our lemma). Then this implies the problem. $\square$

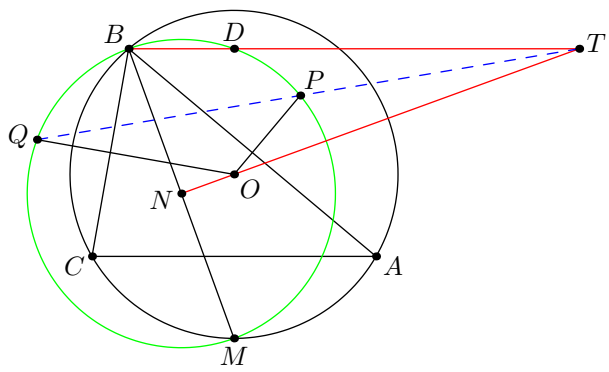
§1.1 Demonstrations

Consider the following problem:

Example 1.3 (Shortlist 2014 G3)

Let Ω and O be the circumcircle and the circumcentre of an acute-angled triangle ABC with $AB > BC$. The angle bisector of $\angle ABC$ intersects Ω at $M \neq B$. Let Γ be the circle with diameter BM. The angle bisectors of $\angle AOB$ and $\angle BOC$ intersect Γ at points P and Q, respectively. The point R is chosen on the line PQ so that $BR = MR$. Prove that $BR \parallel AC$.

Our solution:



We use part of the lemma above, as well as the following:

Claim — Consider a circle Ω with four points A, B, C and D with M the midpoint of AB. Then $(A, B; C, D) = -1$ if and only if line AB is an angle bisector of CMD.

Proof. This is actually true from EGMO, but the if and only if is clearer here. Let the center of Ω be O, and WLOG $\angle CMA$ is acute. Then $\angle CMA = \angle DMA$ just implies that CM and DM are reflections along OM, hence if E is the interssection of CM with Ω then we find that $DE \parallel AB$; thus CD must be a symmedian. $\square$

Then back to the problem, let D be the intersection of OM with Γ at $D \neq M$, so then $\angle BDM$ is right implying $BD \parallel AC$. Let $T = BM \cap ON$, and since $MO \perp TB$ and $TO \perp MB$ then O is the orthocenter of TMB.

Let R and S be the tangents from T to Γ so clearly $RS \parallel BM$ since $RM \perp TN$. Then since O is the orthocenter of TMB,

$$ON \cdot OT = OD \cdot OM \implies O \in \mathrm{RadAx}(\Gamma, (TN))$$

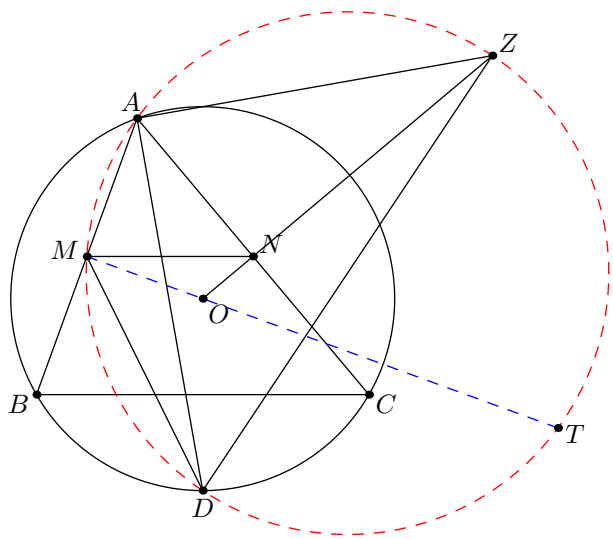
where RadAx is short for Radical Axis. Then it follows that $\boxed{O \in RS}$, since RS is precisly the radical axis since (NT) is orthogonal to Γ.

Now, using our first lemma, we know that RS bisects $\angle POQ$. But then by our second lemma, it must be that $QRPS$ is harmonic. Hence QP passes through the pole of RS, which is T, as desired. $\blacksquare$

Our next example is quick with our lemma.

Example 1.4 (APMO 2017/2)

Let ABC be a triangle with $AB < AC$. Let D be the intersection point of the internal bisector of angle BAC and the circumcircle of ABC. Let Z be the intersection point of the perpendicular bisector of AC with the external bisector of angle $\angle BAC$. Prove that the midpoint of the segment AB lies on the circumcircle of triangle ADZ.



The idea is to take advantage of the symmetry by reflecting Z across the perpendicular bisector of AD to a point T; then T lies on OM. But since $\angle DAZ = 90$, then $AZTD$ is also a rectangle. And as another surprise, $\angle TMA = 90$, so M lies on $(AZTD)$!

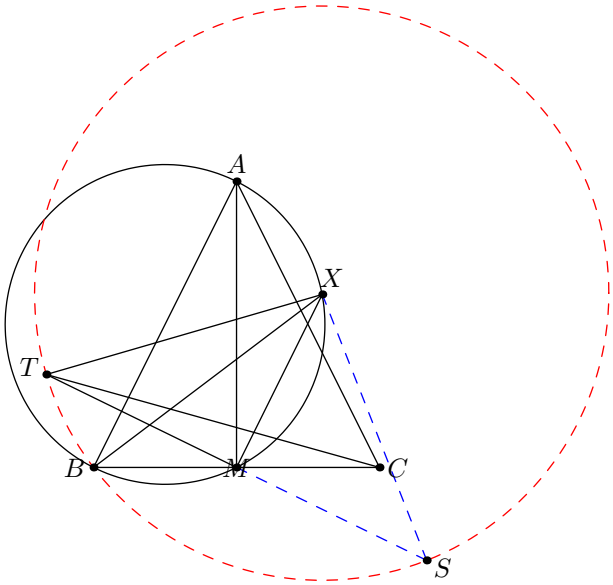
§2 Midpoints and right angles

This section might be somewhat shallow, but the following questions are still quite related.

Example 2.1 (Shortlist 2007 G2)

Denote by M midpoint of side BC in an isosceles triangle $\triangle ABC$ with $AC = AB$. Take a point X on a smaller arc $\widehat{MA}$ of circumcircle of triangle $\triangle ABM$. Denote by T point inside of angle BMA such that $\angle TMX = 90^\circ$ and $TX = BX$.

Prove that $\angle MTB - \angle CTM$ does not depend on choice of X.



There are probably a few facts that seem hard to utilize; namely, the fact that $XB = XT$ and $\angle XMT = 90$. However, the first motivates finding another point S with $XB = XT = XS$, and the second fact motivates to set S to be the reflection of T across M.

But then since M is the midpoint of BC as well, we find that $TCSB$ is a parallelogram, along with the fact that X is the circumcenter of TBS by construction. Angle chasing,

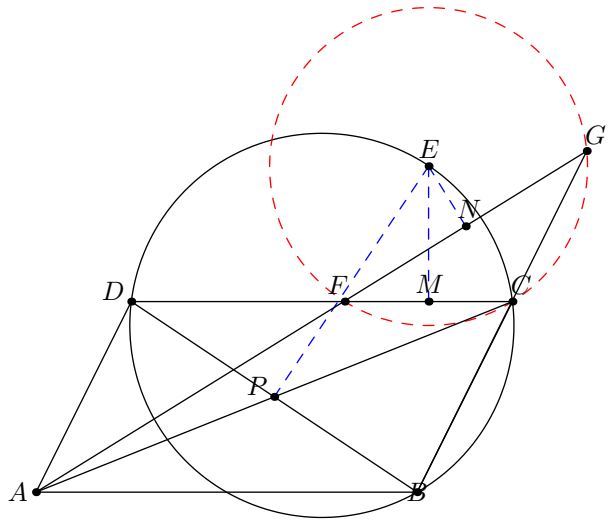
$$\angle BTM - \angle CTM = \frac{1}{2}\angle BXS - \angle BST = \frac{1}{2}\angle BXS - \frac{1}{2}\angle BXT$$

but this is just equal to $\angle MXB = \angle MAB$. ■

Another example is IMO 2007, Problem 2:

Example 2.2 (IMO 2007/2)

Consider five points A, B, C, D and E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let ℓ be a line passing through A. Suppose that ℓ intersects the interior of the segment DC at F and intersects line BC at G. Suppose also that $EF = EG = EC$. Prove that ℓ is the bisector of angle DAB.

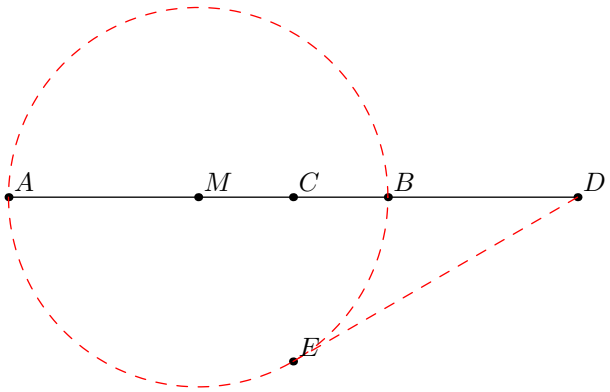


From some angle chasing, we find that showing ℓ is the angle bisector is equivalent to showing $CF = CG$; more angle chasing shows this is also equivalent to showing $EB = ED$.

So it seems like we have a lot of equivalent lengths at E. Isosceles triangles inspire us to drop altitudes to form midpoints; thus, we draw in M, N and P. But then MNP is a Simson line; and in particular, by a homothety, P is the midpoint of BD! Hence $EB = ED$ as needed. ■

§3 Harmonic bundle lemma

Projective geometry is mostly about moving harmonic bundles around. The next lemma really gives more power to harmonic bundles by relating them to a Power of a Point like statement (pun intended).



Lemma 3.1 (Harmonic bundle power of point)

Suppose that A, C, B and D are points all lying on the same line such that $(AB; CD) = -1$. Let the midpoint of AB be M.

Then we have $MA^2 = MC \cdot MD$, as well as $DA \cdot DB = DC \cdot DM$.

Proof. Just draw in the circle with diameter AB. Then since C lies on the polar of D, it follows that $MA^2 = MC \cdot MD$ by inversion. Furthermore, if E is a point on (AB) such

that DE is tangent to (AB), then $EC \perp AB$ and so

$$DA \cdot DB = DE^2 = DC \cdot DM$$

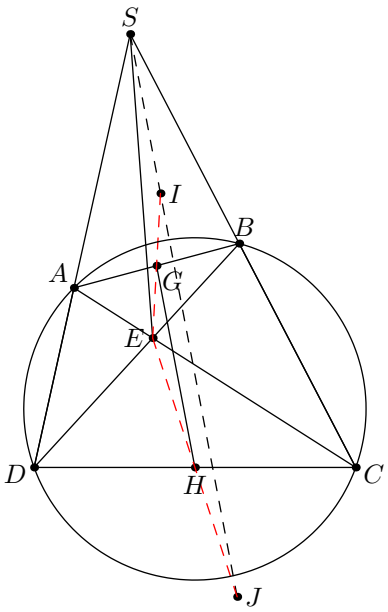
just from the right triangles. □

This is actually quite useful in conjunction with PoP. See the following problem:

Problem 3.2 (Shortlist 2008 G2)**.** Given trapezoid $ABCD$ with parallel sides AB and CD, assume that there exist points E on line BC outside segment BC, and F inside segment AD such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF, and by J the point of intersection of AB and EF. Let K be the midpoint of segment EF, assume it does not lie on line AB. Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ.

§4 Complete quadrilateral isogonalities

It is advised to learn the statement about the first isogonality lemma. It can be found in https://web.evanchen.cc/handouts/GeoSlang/GeoSlang.pdf. We now extend it to find a cute lemma.



The claim is the following:

Lemma 4.1 (Midline isogonality)
Let $ABCD$ be a cyclic quadrilateral and let AD and BC meet at S. Let AC and BD intersect at E. Suppose that G and H are the midpoints of AB and CD respectively. Then the line GH is parallel to the isogonal of SE wirth respect to $\triangle FCD$.

Proof. If we define I and J in the diagram, then applying First Isogonality Lemma on parallelograms $AEBI$ and $CEDJ$ shows that line IJ must actually be the isogonal of SE. But $GH \parallel IJ$, as desired. □

This can be used with the following two problems:

Problem 4.2 (USA TST 2000/2)**.** Let $ABCD$ be a cyclic quadrilateral and let E and F be the feet of perpendiculars from the intersection of diagonals AC and BD to $\overline{AB}$ and $\overline{CD}$, respectively. Prove that $\overline{EF}$ is perpendicular to the line through the midpoints of $\overline{AD}$ and $\overline{BC}$.

This next problem also utilized the harmonic bundle lemma from earlier.

Problem 4.3 (Shortlist 2009 G4)**.** Given a cyclic quadrilateral $ABCD$, let $E = \overline{AC} \cap \overline{BD}$, $F = \overline{AD} \cap \overline{BC}$. The midpoints of $\overline{AB}$ and $\overline{CD}$ are G and H, respectively. Show that $\overline{EF}$ is tangent at E to the circle through the points E, G, and H.